

How exactly a sentence/para $\rightarrow$ 1 vector?

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4# Text embeddings vs. Token embeddings

1)

The sentence can be represented by 2 different ways
e.g. "The dog is sleeping"

case i) $['The', 'dog', 'is', 'sleeping', '.']$

$v_1 \quad v_2 \quad v_3 \quad v_4 \quad v_5$

so we have multiple vectors (Represents each token)

case ii) "The dog is sleeping" $\Rightarrow$ One vector

2) Token embedding

We know,

$$E \in \mathbb{R}^{V \times d}$$

suppose, $V = 50000$ and $d = 768$

then we have 50000 rows $\times$ 768 nos. per row

If "dog" $\rightarrow$ token id 17
then, $E[17] \in \mathbb{R}^{768}$

{Only 1 token ($V=1$)}

Conceptually,

"dog" $\rightarrow 17 \rightarrow E[17] \rightarrow [0.21, -0.44, 0.73, \dots]$

This is a token level embedding/representation.

3) These vectors holds no meaning.

4)

Consider the sentence

"The dog is sleeping."

$['The', 'dog', 'is', 'sleeping', '.'] \rightarrow 5 \text{ tokens}$

h_1

h_2

h_3

h_4

h_5

$\Rightarrow$ Contextual
vectors of each
token

Token embedding : $\rightarrow$ token
 $\downarrow$
vector

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After processing through X_{vec} , we conceptually have:

$$H = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \end{bmatrix}$$

Instead of writing h_1, h_2, h_3, h_4, h_5 every time, we can put them together in 1 matrix:

So,

H = collection of all token representations

If 'n' tokens and each vector has d dimensions.

$$H \in \mathbb{R}^{n \times d}$$

$$\text{Here, } H \in \mathbb{R}^{5 \times 768}$$

So, H is a collective sequence of contextual token vectors. It is not yet one sentence embedding.

5) But RAG requires 1 vector

6) Text embedding

A text embedding is a fixed-dimensional vector representing a larger text span.

That text span could be

— a sentence, a paragraph, a doc. chunk, a query

for ex. "The dog is sleeping on the sofa."

$\downarrow$
Embedding model

$$z = [0.12, -0.44, 0.73, \dots]$$

$$\text{So, } z \in \mathbb{R}^d$$

Text embedding
Used for

- Semantic search
- RAG
- Retrieval
- clustering
- recommendation

Text embedding flow

many tokens

$\downarrow$
model process their relationships

$\downarrow$
one vector

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* Then how do we get 1 vector for the whole sentence?

So, we need 'z' from 'H'.

Eventually, some embedding mechanism transforms H into $z \in \mathbb{R}^{d'}$

i.e.

$$\text{Text} \longrightarrow H \longrightarrow z$$

we know, $H \in \mathbb{R}^{n \times d}$ and $z \in \mathbb{R}^{d'}$

$n \rightarrow$ no of tokens

$d \rightarrow$ internal representation size (dimension of each token vector)

$d' \rightarrow$ final embedding dimensions.

conceptually:

$$H \longrightarrow \boxed{\text{aggregation / pooling / projection}} \longrightarrow z$$

$\Downarrow$
Some mechanism

The exact mechanism depends on the embedding model.

Possible strategies can include things like $\rightarrow$

- Pooling token representations.
- Using a designated representation
- Projection layers
- architectures specifically trained for text embeddings.

for ex $\rightarrow$ $H = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ \vdots \\ h_n \end{bmatrix}$

but we want one vector 'z'.

One possible strategy is to combine the token representations.

say, a simple average:

$$z = \frac{1}{n} \sum_{i=1}^n h_i$$

This is called mean pooling. //

This is just an example, real embedding models can use more sophisticated architectures and objectives.

* Overall concept

- Imagine text: "Employees receive 20 days of paid annual leave."

Token-level representations:

$$h_1, h_2, h_3, h_4, \dots, h_g$$

Then embedding system produces:

One final vector with dimension $(d) = 768$ (assumption)

$$\text{so, } d_{\text{chunk}} \in \mathbb{R}^{768}$$

This what we can store in a vector database.

- Similarly, query works the same way

User asks: "How many vacations days do employees get?"

The query also goes through the embedding model.

$$\text{result: } q \in \mathbb{R}^{768}$$

- Now we have q and $d_1, d_2, d_3, \dots$

$[d_i \rightarrow \text{document chunking}]$

then we calculate cosine similarity

$$\text{sim}(q, d_i)$$

and retrieve the most relevant chunk

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* The most important RAG distinction

When we say:

"We embedded the document."

What we usually mean in a RAG system is:

"We converted a document chunk into a single vector representation suitable for retrieval."

Generally, we are not talking about storing every individual token embedding in the Vector DB.

The database might look conceptually like:

ID	Text chunk	Embedding
1	Employees receive 20 days...	d_1
2	Health insurance is provided...	d_2
3	Parking is available...	d_3

The vector DB indexes the embeddings.

But importantly, it also keeps the original chunk text and metadata so that once d_1 is retrieved, we can get:

"Employees receive 20 days of paid annual leave."
and give it to the LLM.

In short →

Embedding matrix / vocabulary lookup

E



Initial representation of
one token

$E[\text{token_id}]$



Contextual representation of all tokens
in current text

$H[h_1, h_2, h_3, \dots, h_n]$

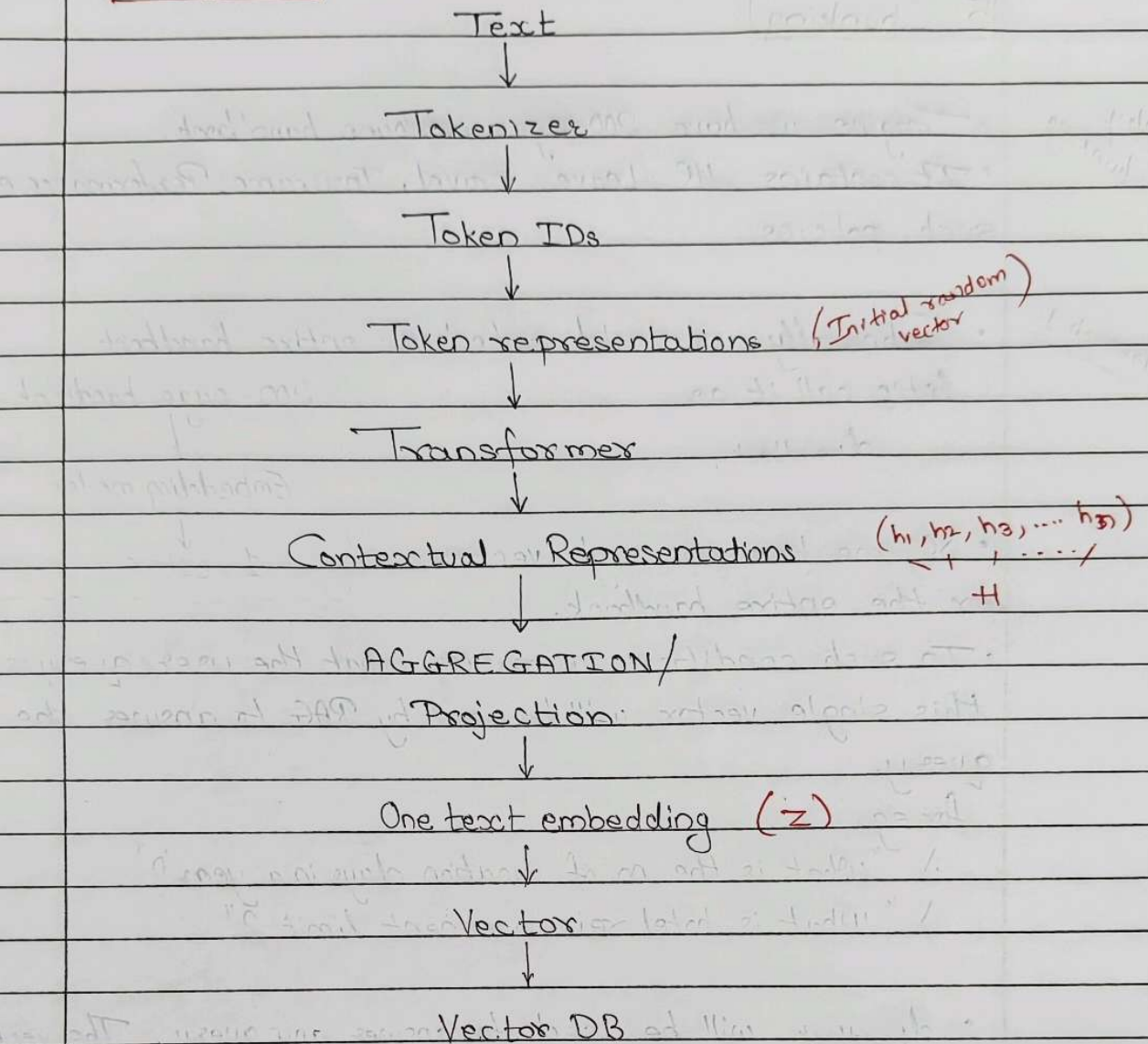


One final text embedding

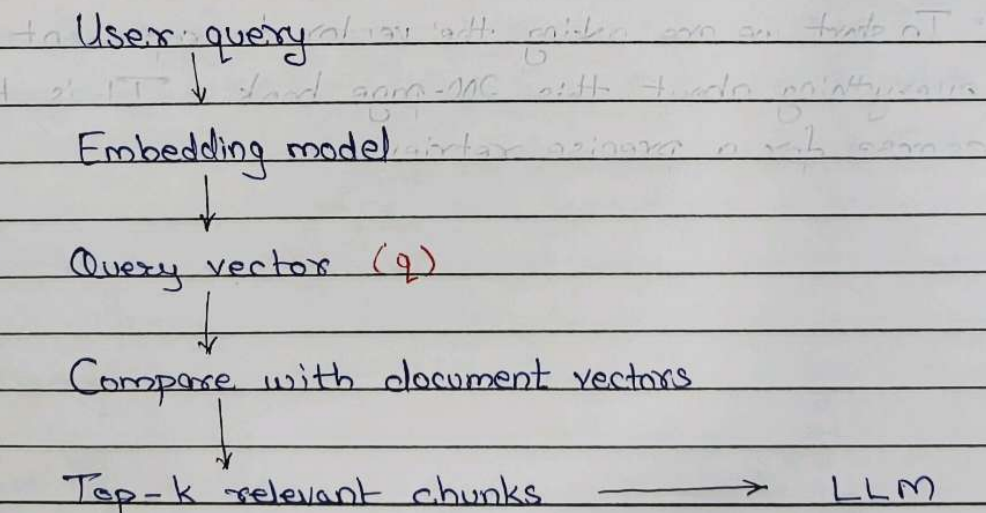
Z

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Full picture

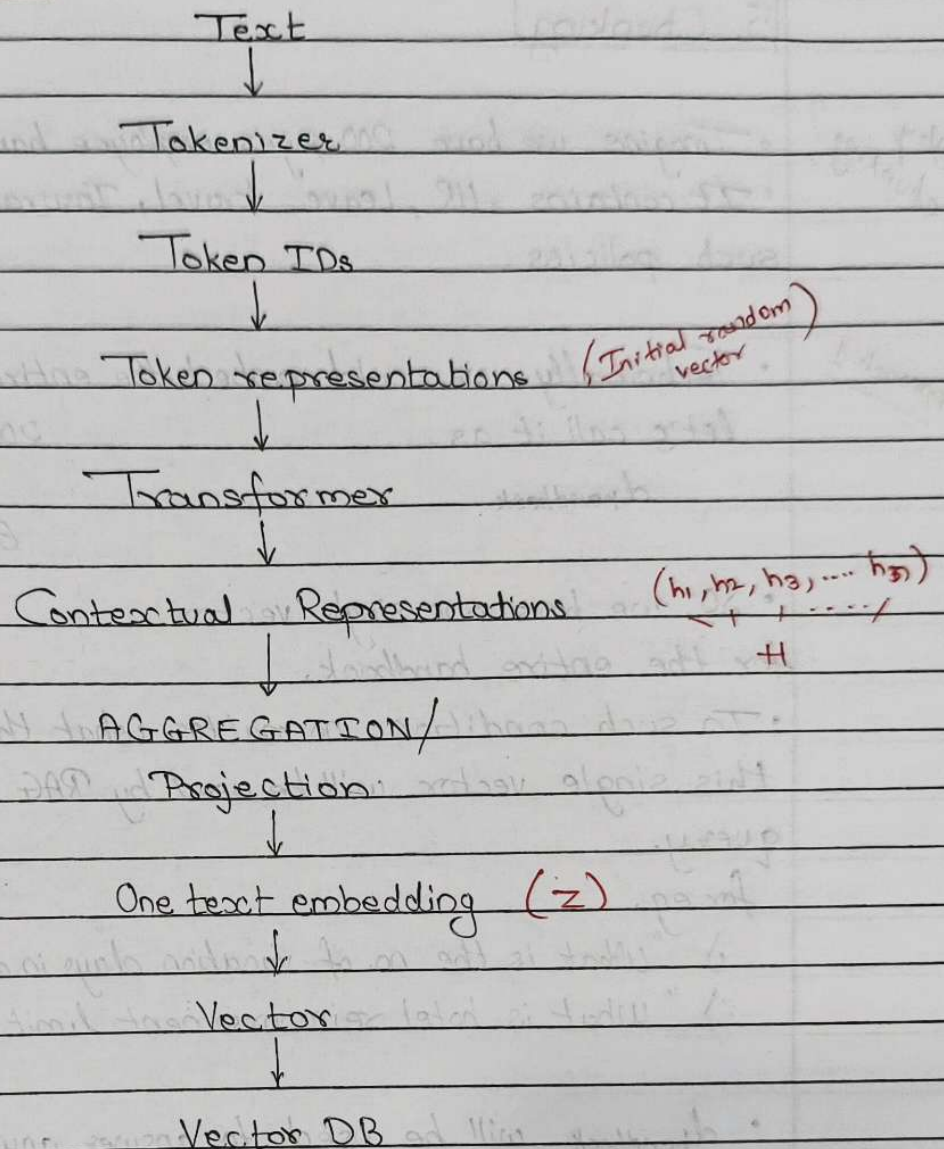


For a RAG Query



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Full picture



For a RAG Query

